# Project Description: Multigraded Homological Algebra and Geometry

This proposal involves research in algebraic geometry and commutative algebra with several connections to computation and combinatorics, as well as a wide array of broader impacts.

- § 1 **Results of Prior NSF Support.**
- § 2 **The Geometry of Multigraded Syzygies on Stacks**
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## 1. Results of Prior NSF Support

During 2020-2022 the PI was supported by an NSF Postdoctoral Research Fellowship (NSF Grant No. MSPRF DMS-2002239, $150,000) titled *Asymptotic Syzygies in Algebraic Geometry.* In this period the PI made major contributions to extending our understanding of homological algebra on toric varieties (Section 1.2) and the cohomology of moduli spaces (Section 1.4). These works resulted in 6 papers being posted to the arXiv [23, 28, 27, 16, 15, 4].

During 2015-2020 the PI was supported by an NSF Graduate Research Fellowship (NSF Grant No. DGE-1256259, $150,000) title *Syzygies in Algebraic Geometry.* In this period the PI made substantial contributions to commutative algebra and algebraic geometry, including furthering our understanding of syzygies on higher dimensional varieties. This resulted in the PI posting 8 new papers to the arXiv [1, 14, 24, 29, 25, 26, 22, 20].

Of the 14 new papers the PI posted to the arXiv during these periods of support 10 have been accepted for publication, including in: *Algebra & Number Theory* [24], *Geometry & Topology* [15], *Journal of Algebra* [23], and *Experimental Mathematics* [25]. The PI contributed to 4 open-source software packages, one public-facing database, and two articles for the *Notices of the AMS* [11, 21]. A detailed summary of these results and their intellectual merits continues in the following sections.

**1.1 Asymptotic Syzygies** Broadly, asymptotic syzygies is the study of the graded Betti numbers (i.e. the syzygies) of a projective variety as the positivity of the embedding grows. For a smooth projective variety $X \subset \mathbb{P}^{n_d}$ embedded by a very ample line bundle $L_d$; following [49], set,

$$\rho_q(X, L_d) \;\; := \;\; \frac{\#\left\{p \in \mathbb{N} \mid \, \big| \, \beta_{p,p+q}(X, L_d) \neq 0\right\}}{n_d},$$

which is the percentage of non-zero syzygies appearing in row $q$ of the Betti table [47, Theorem 1.1]. The asymptotic perspective asks how $\rho_q(X; L_d)$ behaves along the sequence of line bundles $(L_d)_{d\in\mathbb{N}}$. In this language, Green's work on the syzygies of high degree curves can be phrased as follows.

**Theorem 1.1.** [61] *Let $X \subset \mathbb{P}^n$ be a smooth projective curve. If $(L_d)_{d\in\mathbb{N}}$ is a sequence of very ample line bundles on $X$ such that* $\deg L_d = d$ *then* $\lim_{d\to\infty} \rho_2(X; L_d) = 0$.

Put differently, asymptotically the syzygies of curves are as simple as possible, occurring in the lowest possible degree. Ein and Lazarsfeld established the opposite asymptotic behavior for higher dimensional varieties under the analogous positivity condition.

**Theorem 1.2.** [46, Theorem C] *Let $X \subset \mathbb{P}^n$ be a smooth projective variety,* $\dim X \geq 2$*, and fix an index* $1 \leq q \leq \dim X$*. If $(L_d)_{d\in\mathbb{N}}$ is a sequence of very ample line bundles such that $L_{d+1} - L_d$ is constant and ample then* $\lim_{d\to\infty} \rho_q(X; L_d) = 1$.

My work weakens the ample-growth hypothesis to semi-ample growth. Recall that a line bundle $L$ is *semi-ample* if $|kL|$ is base point free for $k \gg 0$ (e.g. $\mathcal{O}(1,0)$ on $\mathbb{P}^n \times \mathbb{P}^m$). I established the following asymptotic nonvanishing result for $PP^n \times \mathbb{P}^m$ embedded by $\mathcal{O}(d_1, d_2)$.

**Theorem 1.3.** [22, Corollary B] *Let $X = \mathbb{P}^n \times \mathbb{P}^m$ and fix an index $1 \leq q \leq n+m$. There exist constants $C_{i,j}$ and $D_{i,j}$ such that*

$$\rho_q\left(X; \mathcal{O}\left(d_1, d_2\right)\right) \geq 1 - \sum_{\substack{i+j=q \\ i \leq n,\, j \leq m}} \left( \frac{C_{i,j}}{d_1^i d_2^j} + \frac{D_{i,j}}{d_1^{n-i} d_2^{m-j}} \right) - O\left( \begin{matrix} \text{lower ord.} \\ \text{terms} \end{matrix} \right).$$

If both $d_1 \to \infty$ and $d_2 \to \infty$ then $\rho_q\left(\mathbb{P}^n \times \mathbb{P}^m; \mathcal{O}(d_1, d_2)\right) \to 1$, recovering the results of Ein and Lazarsfeld for $\mathbb{P}^n \times \mathbb{P}^m$. However, if $d_1$ is fixed and $d_2 \to \infty$ (i.e. semi-ample growth) my results bound the asymptotic percentage of non-zero syzygies away from zero. In fact I conjecture that, unlike in previously studied cases, in the semi-ample setting $\rho_q\left(\mathbb{P}^n \times \mathbb{P}^m; \mathcal{O}(d_1, d_2)\right)$ does not approach 1. Proving this would require a vanishing result for asymptotic syzygies, which is open even in the ample case [46, Conjectures 7.1, 7.5].

**1.2 Eisenbud-Goto on Products of Projective Space** The Castelnuovo–Mumford Regularity of a projective variety $X \subset \mathbb{P}^r$ is a measure of the complexity of $X$ given in terms of the vanishing of certain cohomology groups of $X$. Roughly speaking one should think about Castelnuovo–Mumford regularity as being a numerical measure of geometric complexity. Eisenbud and Goto showed that regularity is closely connected to interesting homological properties.

**Theorem 1.4.** [48] *Let $\mathcal{F}$ be a coherent sheaf on $\mathbb{P}^n$ and $M = \bigoplus_{e \in \mathbb{Z}} H^0(\mathbb{P}^n, \mathcal{F}(e))$ the corresponding section module. The following are equivalent: (1) $M$ is $d$-regular; (2) $\beta_{p,q}(M) = 0$ for all $p \geq 0$ and $q > d + i$; (3) $M_{\geq d}$ has a linear resolution.*

A toric variety $Y$ is a well-behaved compactification of a torus, which under minor hypotheses, can be written as a quotient $Y = [(\mathbb{C}^n \setminus \mathbb{V}(B))/(\mathbb{C}^\times)^r]$ where $B$ is the *irrelevant ideal* of $Y$; for $Y \cong \mathbb{P}^n$ the irrelevant ideal $B = \langle x_0, \ldots, x_n \rangle$. Cox proved a beautiful correspondence between coherent sheaves on $Y$ and $\mathbb{Z}^r$-graded ($B$-saturated) modules over a $\mathbb{Z}^r$-graded polynomial ring $S = \mathbb{C}[x_0, \ldots, x_n]$, which is known as the Cox ring of $Y$ [36]. This yields a rich dictionary between *multigraded* homological algebra over $S$ and the geometry of $Y$. Maclagan and Smith defined multigraded Castelnuovo–Mumford regularity on toric varieties in terms of the vanishing of cohomology groups, Let $Y$ be a smooth projective toric variety so that $\operatorname{Pic}(Y) \cong \mathbb{Z}^r$ and the nef cone of $Y$ is a full-dimensional cone of $\operatorname{Pic}(Y)_{\mathbb{R}}$. For $\mathbf{v} \in \mathbb{Z}^r$ set $|\mathbf{v}| := v_1 + v_2 + \cdots v_r$.

**Definition 1.5.** [92, Definition 6.1] *With the set-up as in the previous paragraph if $\mathcal{F}$ is a coherent module on $X$ then we say that $\mathcal{F}$ is* $\mathbf{d}$-regular *for $\mathbf{d} \in \mathbb{Z}^r$ if and only if $H^i(Y, \mathcal{F}(\mathbf{e})) = 0$ for all $i > 0$ and all $\mathbf{e} \in L_i(\mathbf{d})$ where*

$$L_i(\mathbf{d}) := \bigcup_{\mathbf{v} \in \mathbb{Z}^r, |\mathbf{v}| = i} \left(\mathbf{d} - \mathbf{v} + \operatorname{Nef}(X)\right).$$

*We then define the* multigraded regularity *of $\mathcal{F}$ to be* $\operatorname{reg}(\mathcal{F}) := \{\mathbf{d} \in \mathbb{Z}^r \mid \mathcal{F}$ *is* $\mathbf{d}$*-regular*$\}$.

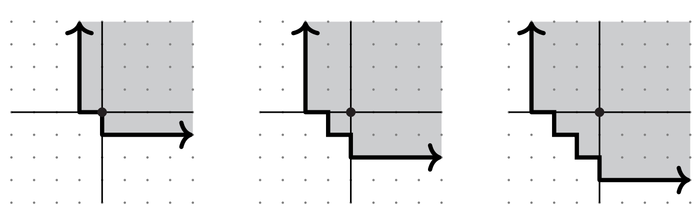

FIGURE 1. The regions $L_1(0,0)$, $L_2(0,0)$, and $L_3(0,0)$ for $Y = \mathbb{P}^n \times \mathbb{P}^m$, in which case $\operatorname{Pic}(Y) \cong \mathbb{Z}^2$ and $\operatorname{Nef}(Y)$ is the first quadrant.

We can define the multigraded regularity of a $\mathbb{Z}^r$-graded module $M$ over $S$ in terms of analogous vanishing of local cohomology modules $H^I_B(M)$ where $B$ is the irrelevant ideal of $Y$. The multigraded Castelnuovo–Mumford regularity is not a single number, but an infinite subset of $\mathbb{Z}^r$.

The obvious approaches to generalize Theorem 1.4 to a product of projective spaces fail. For example, the multigraded Betti numbers do not determine multigraded Castelnuovo–Mumford regularity [28, Example 5.1]. Despite this we show that part (3) of Theorem 1.4 can be generalized. To do so we introduce the following generalization of linear resolutions.

**Definition 1.6.** *A complex $F_\bullet$ of $\mathbb{Z}^r$-graded free $S$-modules is $\mathbf{d}$-quasilinear if and only if $F_0$ is generated in degree $\mathbf{d}$ and each twist of $F_i$ is contained in $L_{i-1}(\mathbf{d}-\mathbf{1})$.*

**Theorem 1.7.** [28, Theorem A] *Let $M$ be a (saturated) finitely generated $\mathbb{Z}^r$-graded $S$-module:*

$$M \text{ is } \mathbf{d}\text{-regular} \iff M_{\geq \mathbf{d}} \text{ has a } \mathbf{d}\text{-quasilinear resolution.}$$

The proof of Theorem 1.7 is based on a spectral sequence argument relating the Betti numbers of $M_{\geq \mathbf{d}}$ to the Fourier–Mukai transform of $\widetilde{M}$ with Beilinson's resolution of the diagonal as the kernel. This result provides an effective way to compute whether a module is regular at a given multidegree $\mathbf{d} \in \mathbb{Z}^r$ and construct so called "short virtual resolutions" on products of projective spaces.

**1.3 Asymptotic Multigraded Regularity of Ideals** Building upon our work discussed above – and inspired by previous work on $\mathbb{P}^n$ [10, 41, 86] – I studied the asymptotic regularity of powers of ideals on arbitrary toric varieties. My collaborators and I show that the multigraded regularity of powers of ideals is bounded and translates in a predictable way. In particular, the regularity of $I^t$ essentially translates within $\mathrm{Nef}(Y)$ in fixed directions at a linear rate.

**Theorem 1.8.** [27, Theorem 4.1] *There exists a degree $\mathbf{a} \in \mathrm{Pic}(Y)$, depending only on $I$, such that for each integer $t > 0$ and each pair of degrees $\mathbf{q}_1, \mathbf{q}_2 \in \mathrm{Pic}(Y)$ satisfying $\mathbf{q}_1 \geq \deg f_i \geq \mathbf{q}_2$ for all generators $f_i$ of $I$, we have*

$$t\mathbf{q}_1 + \mathbf{a} + \operatorname{reg} S \subseteq \operatorname{reg}\big(I^t\big) \subseteq t\mathbf{q}_2 + \mathrm{Nef}(Y).$$

**1.3.1 Syzygies via Highly Distributed Computing** It is quite difficult to compute examples of syzygies. For example, until recently the syzygies of the projective plane embedded by the $d$-uple Veronese embedding were only known for $d \leq 5$. My co-authors and I exploited recent advances in numerical linear algebra and high-throughput high-performance computing to generate a number of new examples of Veronese syzygies. A follow-up project used similar computational approaches to compute the syzygies of $\mathbb{P}^1 \times \mathbb{P}^1$ in over 200 new examples. This data provided support for several existing conjectures, as well as led us to make a number of new conjectures [25, 23]. The data is publicly available via SyzygyData.com and a package for Macaulay2 [24, 60].

**1.4 Top-Weight Cohomology of $\mathcal{A}_g$** The moduli stack of abelian varieties of dimension $g$, $\mathcal{A}_g$, is the smooth DM stack $[\mathbb{H}_g/\mathrm{Sp}(2g,\mathbb{Z})]$ parameterizing isomorphism classes of principally polarized abelian varieties of dimension $g$. Despite extensive study, much of its geometry is unknown, for example its rational cohomology is fully known only for $g \leq 4$ [70, 65, 68, 69] and Grushevsky asked if $H^{2k+1}(\mathcal{A}_g;\mathbb{Q}) \neq 0$ for some $g$ and $k$ [63]. Building on [31], my co-authors and I developed methods to analyze a canonical quotient of $H^k(\mathcal{A}_g;\mathbb{Q})$ and produced new nontrivial classes.

**Theorem 1.9.** [15, Theorem A] *The rational cohomology $H^k(\mathcal{A}_g;\mathbb{Q}) \neq 0$ for:*

$$(g,k) = (5,15), (5,20), (6,30), (7,28), (7,33), (7,37), \text{ and } (7,42).$$

This theorem provided an answer to Grushevsky. Since $\mathcal{A}_g$ is a rational classifying space for $\mathrm{Sp}_{2g}(\mathbb{Z})$ we have $H^*(\mathcal{A}_g;\mathbb{Q}) \cong H^*(\mathrm{Sp}_{2g}(\mathbb{Z});\mathbb{Q})$, and thus, Theorem 1.9 gives new non-vanishing results for $\mathrm{Sp}_{2g}(\mathbb{Z})$. Our approach uses Deligne's mixed Hodge theory and tropical geometry to compute the top-weight cohomology of $\mathcal{A}_g$ for $g \leq 7$. Subsequent work of the PI generalized this to tropicalizing Shimura varieties [4], highlighted in a 2025 SRI in Algebraic Geometry plenary.

### 1.5 Broader Impacts from Prior NSF Support

**1.5.1 Organizing.** I organized 11 conferences: *Math Careers Beyond Academia* (50 participants), *M2@UW* (45 participants), *Geometry & Arithmetic of Surfaces* (40 participants), *CAZoom* (70 participants), *WAGS* (100 participants), *Gems of Combinatorics I & II* (40 & 30 participants), Spec($\overline{\mathbb{Q}}$) (50 participants), *BATMOBILE (2022)* (20 participants), *Gems of Commutative Algebra I & II* (35 & 40 participants). In this time I organized three special sessions at AMS Sectional Meetings and the Joint Math Meetings. Many of these workshops, namely the *Gems of* series, focused on supporting the next generation of researchers by building community and networks.

**1.5.2 Math Circles.** From 2015 through 2018 I was heavily involved in the Madison Math Circle, including 2 years as the lead organizer. As an organizer, I worked to build stronger connections between the Madison Math Circle, local schools, teachers, and other outreach organizations. This led to weekly attendance to more than double. I also created a new outreach arm of the circle, which visits high schools around the state of Wisconsin to better serve students. While a post-doc at UC Berkeley I volunteered as a speaker with the Berkeley Math Circle.

**1.5.3 Mentoring.** While a graduate student and post-doc I led four reading courses with undergraduates in algebraic and geometry, volunteered for *Girls' Math Night Out* to promote mathematics to high schoolers, and mentored 6+ students via the AWM's Mentoring Network. During Summer 2021 I led 6 undergraduates on a research project related to [15]. In Summer 2022 I advised an undergraduate student – now a math Ph.D. student with an NSF GRF – on a research project related to the work in Section 1.2.

**1.5.4 Virtual Mathematics.** In response to the COVID-19 pandemic and the shift of many mathematical activities to virtual formats, I worked to find ways for these online activities to reach those often at the periphery. During Summer and Fall 2020, I helped with Ravi Vakil's *Algebraic Geometry in the Time of Covid* project. This massive online open-access course in algebraic geometry brought together $\sim 2,000$ participants from around the world. In Spring 2021, I organized an 8-week virtual reading course for undergraduates in algebraic geometry and commutative algebra.

**1.5.5 Community.** While a graduate student I co-founded oSTEM@UW as a resource for students in STEM, and organized a peer networking event *qGrads*, which had over 350 members. During 2020-2023 I served on the board of *Spectra*, including as its inaugural president. I oversaw the growth and formalization of the organization, including the creation and adoption of bylaws, the creation of an invited lecture at the JMM, and a fundraising campaign that has raised over $20,000.

## 2. The Geometry of Multigraded Syzygies on Stacks

Let $X$ be a projective variety and $L$ a very ample line bundle. Considering the embedding of $X$ into projective space $X \hookrightarrow \mathbb{P}H^0(X,L) \cong \mathbb{P}^n$ defined by $L$, let $S_X = S/I_X$ where $S = \mathbb{C}[x_0,\ldots,x_n]$ is the standard graded polynomial ring and $I_X$ is the homogeneous defining ideal of $X$. The minimal graded free resolution of $S_X$ as an $S$-module is an exact sequence:

$$0 \longleftarrow S_X \xleftarrow{\epsilon} F_0 \xleftarrow{d_1} \cdots \qquad\qquad \cdots \xleftarrow{d_{k-1}} F_{k-1} \xleftarrow{d_k} F_k \longleftarrow \cdots$$

where $F_p$ is a graded free $S$-module, and the differentials satisfy some minimality conditions. Since $F_p$ is a graded free $S$-module it can be written as $\bigoplus_q S(-q)^{\beta_{p,q}(S_X)}$. The module $S(-q)$ is the ring $S$ with a twisted grading, so that $S(-q)_d$ is equal to $S_{d-q}$ where $S_{d-q}$ is the graded piece of degree $d-q$. The $\beta_{p,q}(S_X)$ are known as the graded Betti numbers (or syzygies) of the pair $(X,L)$, and we often denote them as $\beta_{p,q}(X;L)$ to indicate their dependence not just on $X$ but on the line bundle $L$ as well. That is to say the syzygies depend on the extrinsic embedding of $X$ into $\mathbb{P}^n$. An overarching theme in the study of syzygies in algebraic geometry is understanding ways that minimal graded free resolutions and graded Betti numbers also capture the intrinsic geometry of $X$.

An example of this is Mumford's work showing varieties of high degree are defined by quadrics [96]. The connection between the geometry of $X$ and its minimal graded free resolutions has been made most precise when $X$ is a curve. For example, consider the following classical theorem of Noether and Petri concerning canonical curves.

**Theorem 2.1** (Noether-Petri Theorem)**.** *Let $X$ be a smooth projective curve of genus $g$. If $X$ does not have a degree 2 cover of $\mathbb{P}^1$ then the canonical bundle $K_X$ defines a projectively normal embedding of $X$ into $\mathbb{P}^{g-1}$. Further, if $X$ does not admit a degree 3 cover of $\mathbb{P}^1$ then the image of $X \subset \mathbb{P}^{g-1}$ is defined by quadrics.*

In the language of minimal graded free resolutions this theorem can be translated as follows: i) If $X$ does not admit a cover of $\mathbb{P}^1$ of degree 2 then $\beta_{0,0}(X;K_X) = 1$ and $\beta_{0,q}(X;K_X) = 0$ for all $q \neq 0$, and ii) If $X$ does not admit a cover of $\mathbb{P}^1$ of degree 3 then $\beta_{1,q}(X;K_X) = 0$ for all $q \neq 2$. Other results in this vein for curves include Castelnuovo's theorem that if $\deg(L) \geq 2g+1$ then $L$ is very ample and if $\deg(L) \geq 2g+2$ then $I_X$ is generated by quadrics, and the theorem that if $\deg(L) \geq 2g+1$ then $S_X$ is Cohen-Macaulay.

**2.1 $N_p$-Conditions for Stacky Curves** A *stacky curve* is a smooth proper geometrically connected Deligne-Mumford stack of dimension 1 which contains a dense open subscheme. Effectively a stacky curve is a curve with a finite number of special points – called stacky points – which have "fractional degrees". Note this intuition can be made precise in the sense that any stacky curve is (Zariski) locally the quotient of a smooth affine curve by a finite group [103, Lemma 5.3.10]. For simplicity we work over $\mathbb{C}$, however, everything remains true over an arbitrary field if one places minor conditions on the stacky curve.

In [103] Voight and Zureick-Brown develop a theory of line bundles on stacky curves by transferring properties of divisors from their smooth coarse spaces. The key thing to note is that the degree of a line bundle on a stacky curve may be a rational number. Associated to a stacky curve $\mathcal{X}$ and a line bundle $\mathcal{L}$ the *section module* of $\mathcal{X}$ with respect to $\mathcal{L}$ is $R(\mathcal{X};\mathcal{L}) \coloneqq \bigoplus_{k\in\mathbb{Z}} H^0(\mathcal{X}, k\mathcal{L})$.

A fundamental difference from the classical theory is that $\mathcal{L}$ provides an embedding into a weighted projective stack, not standard projective space [13]. Algebraically this means that $R(\mathcal{X};\mathcal{L})$ is a module over a polynomial ring with a non-standard $\mathbb{Z}$-grading. Assuming $\mathcal{L}$ is sufficiently positive $R(\mathcal{X};\mathcal{L}) \cong S/I$ for a non-standard $\mathbb{Z}$-graded polynomial ring $S = \mathbb{C}[x_1, x_2, \ldots, x_n]$ where $\deg(x_i) = w_i \in \mathbb{Z}$ and $I \subset S$ some homogeneous ideal.

*Example* 2.2. Suppose $\mathcal{X}$ is a stacky curve with two stacky points each having $\mathbb{Z}/2\mathbb{Z}$ as its stabilizer and whose coarse space has genus 1. The canonical ring of $\mathcal{X}$ can be minimally presented as

$$R(\mathcal{X};K_{\mathcal{X}}) \cong \mathbb{C}[x_1,x_2,x_3]/\langle x_3^2x_1 - bx_3x_2^6 - x_1^2 - ax_2^8\rangle$$

where $a, b \in \mathbb{C}$ and $\deg(x_1) = 4$, $\deg(x_2) = 1$, and $\deg(x_3) = 2$ [103, Remark 8.3.6]

**Research Goal 2.3.** *Understand the syzygies of $R(\mathcal{X}, \mathcal{L})$ where $\mathcal{X}$ is a stacky curve and $\mathcal{L}$ is a line bundle on $\mathcal{X}$.*

The heuristic for this goal is two-fold: i) the syzygies of section rings of stacky curves should behave roughly like the syzygies of weighted log complete series on a curves (see [18]), and ii) Stacky-ness should be thought of as increasing the positivity of line bundle. Justification for the first heuristic comes in the form of the following lemma.

**Lemma 2.4.** *Let $\mathcal{X}$ be a stacky curve with $\pi : \mathcal{X} \to X$ its coarse space. If $\mathcal{L}$ is a line bundle on $\mathcal{X}$ then there exists a weighted log incomplete series $W$ on $X$ such that $R(\mathcal{X}; \mathcal{L}) \cong \bigoplus_{d \in \mathbb{Z}} H^0(X, dW)$.*

Justification for the second heuristic is less precise, however, if $\mathcal{L}$ is a line bundle on $\mathcal{X}$ then $\deg(\mathcal{L})$ is equal to $\deg(L) + \sum_i (1 - \frac{1}{e_i})$ where $L$ is a line bundle on $X$ and $e_i$ are the order of the stabilizers of the stacky points. Thus, $\deg(\mathcal{L})$ is "increased" by the presence of stacky points.

**Research Problem 2.5.** *Let $\mathcal{X}$ be a stacky curve with $e$ the maximum order of the stabilizer groups of stacky points on $\mathcal{X}$. Let $\mathcal{L}$ a line bundle on $\mathcal{X}$.*

*(1) Find a characterization in terms of $\deg(\mathcal{L})$ for when $R(\mathcal{X} : \mathcal{L})$ is Cohen-Macaulay.*
*(2) Find a characterization in terms of $\deg(\mathcal{L})$ for when $R(\mathcal{X} : \mathcal{L})$ is generated in degrees$\leq 2e$.*

Brown and Erman recently introduced an analog of Green's $N_p$ conditions for varieties embedded in weighted projective space [18, Definition 1.2]. Given this, I would like to understand which $N_p$ properties stacky curves satisfy.

**Research Problem 2.6.** *Let $\mathcal{X}$ be a stacky curve whose coarse space has genus 0 or 1. If $\mathcal{L}$ is a line bundle on $\mathcal{X}$ and $\deg(\mathcal{L}) \gg 0$, what $N_p$ conditions does $R(\mathcal{X}; \mathcal{L})$ satisfy? Determine precise bounds on $\deg(\mathcal{L})$ guaranteeing $N_p$.*

We conjecture that $R(\mathcal{X}; \mathcal{L})$ satisfies higher $N_p$ conditions that grow roughly linearly with $\deg(\mathcal{L})$. In ongoing work with Maya Banks and Christopher Manon, we are generalizing Green's kernel bundle techniques [61] to prove this and resolve Problem 2.6. We have successfully constructed a multigraded generalization of these bundles whose cohomology computes the multigraded Betti numbers of $R(\mathcal{X}; \mathcal{L})$. Our current objective is to prove the necessary vanishing theorems for these bundles, which would establish the conjecture; even just under the assumptions the the coarse space of $\mathcal{X}$ has genus 0 or 1.

**2.2 Green's Conjecture for Canonical Stacky Curves** One perspective on the Noether-Petri Theorem is that the syzygies of a smooth projective curve $X$ seem to be closely related to the degrees for which $X$ admits a cover of $\mathbb{P}^1$. This heuristic turns out to be true in several ways, but to understand the higher syzygies of canonical curves one needs a slightly more subtle invariant. The Clifford index of a smooth curve $X$ is defined to be:

$$\operatorname{Cliff}(X) := \min \left\{ \deg(L) - 2\dim H^0(X, L) + 2 \quad \middle| \begin{matrix} L \in \operatorname{Pic}(X) \\ \deg(L) \leq g-1 \quad \dim H^0(X, L) \geq 2 \end{matrix} \right\}.$$

As a vast generalization of the Noether-Petri theorem, Green's conjecture states that the syzygies of canonical curves (i.e. the syzygies of a (non-hyperelliptic) curve embedded by the complete linear series of the canonical divisor) are controlled by the Clifford index.

**Conjecture 2.7** (Green's Conjecture [61])**.** *If $X$ is a smooth projective curve then:*

$$\beta_{p,p+2}(X; K_X) = 0 \qquad \textit{for all} \qquad p < \operatorname{Cliff}(X).$$

Stated differently Green's conjecture say that one can read the Clifford index of a smooth curve (which is an intrinsic invariant of the curve, not dependent on the embedding) from the vanishing of the syzygies of the canonical embedding of the curve (which are extrinsic to the curve, dependent on the choice of the embedding). Breakthrough work of Voisin proved this conjecture for general curves of every genus [104, 105], and in recent years several new proofs have been found [3, 81]. Substantial work has gone into finding refinements and extensions of Green's Conjecture [52, 33, 50, 51, 44, 82].

**Research Goal 2.8.** *Formulate and investigate an appropriate generalization of Green's conjecture describing the vanishing of the syzygies of the canonical rings of stacky curves.*

The first steps towards Goal 2.8 were taken by Voight and Zureick-Brown who proved a stacky analog of the Noether-Petri theorem.

**Theorem 2.9.** [103, Theorem 8.4.1] *Let $\mathcal{X}$ be stacky curve whose coarse space has genus $\geq 1$ and let $e$ be the maximum order of the stabilizer groups of the stacky points; then the canonical ring $R(\mathcal{X}; K_{\mathcal{X}})$ is generated by elements in degree at most $3e$ with relations in degree at most $6e$.*

This theorem shows that complexity is bounded by the stacky structure, but these bounds are likely not sharp. For example, when $e = 1$ this fails to recover the classical result, instead allowing both quadrics and cubics. Further, it does not provide a geometric explanation of the failure of $R(\mathcal{X}; K_{\mathcal{X}})$ to be generated in the smallest single degree. We propose to refine Theorem 2.9 by studying the relationship between weighted linear series on $\mathcal{X}$ and the syzygies of the canonical ring of $\mathcal{X}$. A far-reaching goal of this would be to introduce a notion of Clifford index for stacky curves.

**Research Problem 2.10.** *Can Theorem 2.9 be refined by considering the existence of certain stacky or weighted linear series on $\mathcal{X}$? Formulate a stacky analogue of Clifford's index.*

To define a meaningful Clifford index, we need a robust theory of linear series on stacks; including studying the geometry of the stacky Brill-Noether loci $\mathcal{W}^r_d(\mathcal{X})$, parameterizing line bundles of degree $d$ with at least $r + 1$ independent sections, taking the stacky structure into account. The classical Brill-Noether theorem describes the expected dimension, irreducibility and smoothness of these loci for a general curve [62, 53, 57].

**Research Problem 2.11.** *For a general stacky curve $\mathcal{X}$ compute the expected dimension of $\mathcal{W}^r_d(\mathcal{X})$ in terms of the genus of the coarse space and the stacky invariants, and determine whether the stacky Brill-Noether locus is of the expected dimension, smooth away from $\mathcal{W}^{r+1}_d(\mathcal{X})$ and irreducible when positive dimensional.*

Recent work refines classical Brill–Noether theory by tracking data analogous to stacky structure (e.g., specified ramification) [100, 88, 89, 71, 35, 34, 87]. The resulting toolkit – degenerations, limit linear series, and tropical/logarithmic method, etc. – appears well suited to possible adaption to the stacky setting, and I am optimistic that these techniques can be adapted to address Problem 2.11. As an achievable first step (suitable for a graduate student project), we will resolve this question completely for stacky curves whose coarse space has genus 0 or 1 with at most 3 or 4 stacky points, where explicit computation of $\mathcal{W}^r_d(\mathcal{X})$ is more tractable.

Another approach one may take towards Goal 2.8 is by studying the algebraic properties of the canonical rings $R(\mathcal{X}; K_{\mathcal{X}})$. As a step in this direction the PI and co-author has shown that just like classical curves [102, 99], the canonical ring of stacky curves is generally Gorenstein.

**Lemma 2.12.** *If $\mathcal{X}$ is a stacky curve whose coarse space is not hyperelliptic, trigonal, or isomorphic to a plane quintic then the canonical ring $R(\mathcal{X}; K_{\mathcal{X}})$ is Gorenstein.*

The proof of this lemma largely follows the classical argument, after passing to the coarse space, with some slight delicateness needed in handling the non-standard grading. A significantly more subtle algebraic question would be to determine when $R(\mathcal{X}; K_{\mathcal{X}})$ is Koszul, in the sense of Koszulness for modules over a non-standard graded polynomial ring developed recently in [43].

**Research Problem 2.13.** *Let $\mathcal{X}$ be a stacky curve whose coarse space has genus $g$. For what $t$ does the canonical ring $R(\mathcal{X}; K_{\mathcal{X}})$ satisfy the $t$-Koszul property above.*

At this time I have no strong guess for what the correct characterization of when $R(\mathcal{X}; K_{\mathcal{X}})$ is $t$-Koszul should be. However, in exaplace checking $t$-Koszul-ness can be done computationally.

The proof of Theorem 2.9 builds upon the ideas of Schreyer [101] to show the existence of a Gröbner basis for the canonical ring of a stacky curve whose elements have certain degrees. Voight and Zureick-Brown do not explicitly construct such Gröbner bases, however, this should be possible for simple stacky curves. A starting point towards Goal 2.8 is to make Voight and Zureick-Brown's arguments explicit, and then use a computer algebra system like Macaulay2 to compute the minimal graded free resolutions for many examples.

**Research Problem 2.14.** *Compute the minimal graded free resolution of the canonical ring $R(\mathcal{X}, K_{\mathcal{X}})$ for stacky curves $\mathcal{X}$ where the coarse space of $\mathcal{X}$ has genus $\leq 4$, there are at most 4 stacky points, and the stabilizers of the stacky points have order at most 5.*

Building upon code shared by Voight and Zureick-Brown, I have carried out Problem 2.14 for most of the genus zero examples. In many of these computed cases, and as noted in [103, Appendix], the canonical ring is a (weighted) complete intersection. However, there are examples showing that even for small examples the canonical ring of a stacky curve is quite complicated.

*Example* 2.15. Let $\mathcal{X}$ be a stacky curve with three stacky points whose stabilizers are $\mathbb{Z}/4\mathbb{Z}$, $\mathbb{Z}/4\mathbb{Z}$, and $\mathbb{Z}/5\mathbb{Z}$ whose coarse space has genus zero. The canonical ring $R(\mathcal{X}; K_{\mathcal{X}})$ can be presented as $S/I$ where $S = \mathbb{C}[x_1, x_2, x_3, x_4, x_5]$ where $\deg(x_1) = 3$, $\deg(x_2) = \deg(x_3) = 4$, and $\deg(x_4) = \deg(x_5) = 5$ and $I$ is an ideal generated by 7 homogeneous polynomials of degrees $8, 8, 9, 9, 10, 13, 13$. The minimal graded free resolution of $R(\mathcal{X}; K_{\mathcal{X}})$ is shown below (with the degrees suppressed for brevity):

$$0 \longleftarrow R(\mathcal{X}; K_{\mathcal{X}}) \longleftarrow S \longleftarrow S^{\oplus 7} \longleftarrow S^{\oplus 20} \longleftarrow S^{\oplus 30} \longleftarrow S^{\oplus 22} \longleftarrow S^{\oplus 6} \longleftarrow 0.$$

The approach of using large-scale computations to find conjectures concerning syzygies is very much in the same spirit as my prior work on the syzygies of surfaces [25, 23]. While the precise techniques needed to address Problem 2.14 will likely be different, much of the general framework of organizing, distributing, and publicizing such computations will likely be similar.

**2.3 Asymptotic Syzygies of Weighted Projective Stacks** The study of asymptotic syzygies investigates the behavior of Betti numbers $\beta_{p,q}(X, L_d)$ as the positivity of the embedding line bundle $L_d$ grows. Ein and Lazarsfeld showed that for a smooth projective variety $X$, the shape of the Betti table stabilizes as $d \to \infty$ (see Section 1.1). I propose to generalize this theory to smooth DM stacks. Toric stacks, including weighted projective stacks, provide a natural setting for this investigation.

**Research Problem 2.16.** *Let $\mathcal{X}$ be a smooth toric DM stack, $\dim \mathcal{X} \geq 2$, and fix an index $1 \leq q \leq \dim \mathcal{X}$. If $(\mathcal{L}_d)_{d\in\mathbb{N}}$ is a sequence of very ample line bundles such that $\mathcal{L}_{d+1} - \mathcal{L}_d$ is constant and ample compute the limit (if it exists):*

$$\lim_{d\to\infty} \rho_q(\mathcal{X}; \mathcal{L}_d) = \lim_{d\to\infty} \frac{\#\left\{p \in \mathbb{N} \mid \beta_{p,p+q}(\mathcal{X}, \mathcal{L}_d) \neq 0\right\}}{n_d}.$$

We will begin by focusing on specific examples, such as weighted projective planes. Let $\mathcal{P}(a, b, c)$ be the quotient stack $(\mathbb{C}^3 \setminus \{0\})/\mathbb{C}^\times$ where $\mathbb{C}^\times$ acts by $\lambda \cdot (x, y, z) = (\lambda^a x, \lambda^b y, \lambda^c z)$. The coarse space of $\mathcal{P}(a, b, c)$ is the usual weighted projective space $\mathbb{P}(a, b, c)$.

**Research Problem 2.17.** *Fix a positive integer $a > 1$. For what values of $p$ does the weighted projective stacky plane $\mathcal{P}(1, 1, a)$ embedded by $\mathcal{O}(d)$ satisfy or not satisfy $N_p$.*

The stack $\mathcal{P}(1, 1, a)$ has a single stacky point with stabilizer $\mathbb{Z}/a\mathbb{Z}$. Its Cox ring is $S = \mathbb{K}[x, y, z]$ with $\deg(x) = \deg(y) = 1$ and $\deg(z) = a$. We anticipate that the required degree $d$ for $N_p$ to hold will grow linearly with $a$. I hope to approach both Problem 2.16 and Problem 2.17 in a somewhat unified method. I believe it will be possible to adopt the methods of constructing explicit non-zero syzygies in the Koszul cohomology after quotienting by a regular sequence developed in [45] and extended by the PI (see Section 1.1 and [22]). Ideally these can be extended asymptotic results for toric stacks by using a weighted version of Noether normalization argument.

**2.4 Computations with Toric Stacks** The lack of accessible computational tools is a significant bottleneck in the study of homological invariants of stacks. While a general framework is out of reach, toric stacks are tractable: their combinatorics is well developed, they correspond to *stacky fans* – a fan $\Sigma \subset N_{\mathbb{R}}$ and a lattice map $\beta : N \to L$ – that makes computations feasible [54, 55, 12]. I am not

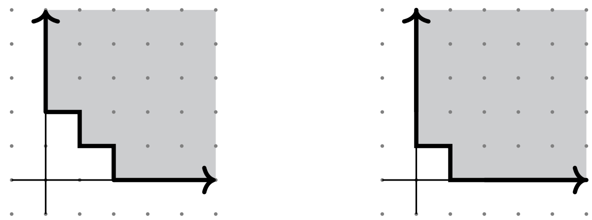

FIGURE 2. Left: The multigraded Castelnuovo–Mumford regularity of $\mathcal{O}_X$ where $X \subset \mathbb{P}^1 \times \mathbb{P}^1$ is the three points $([1:1],[1:4])$, $([1:2],[1:5])$, $([1:3],[1:6])$. Right: The multigraded Castelnuovo–Mumford regularity of $\mathcal{O}_{\mathcal{H}_3}$.

aware of an implementation for working with them. Given the success of the *NormalToricVarieties* package, a *toricStacks* package would benefit both this project and the community.

**Research Problem 2.18.** *Develop a robust software package for Macaulay2 allowing for computations with toric stacks.*

I will develop the Macaulay2 package *toricStacks*, implementing stacky fans and their morphisms, line bundles on toric stacks, Cox rings, and sheaf cohomology. Work began in Summer 2025 with Maya Banks and John Cobb. I have extensive experience building Macaulay2 packages and will involve graduate students. The package will be open-source.

As computations of examples of syzygies have frequently proven fruitful, I would like to extend the database of syzygies I created in [25, 23].

**Research Problem 2.19.** *Adapt the high-performance computing methods in* [25, 23] *to compute the graded Betti numbers of* $\mathcal{P}(1,1,a)$ *embedded by* $\mathcal{O}(d)$ *for* $1 < a \leq 15$ *and* $2 \leq d \leq 6$.

## 3. Castelnuovo–Mumford Regularity: Resolutions, Bounds, and Asymptotic Behavior

Building directly on the PI's work [28, 27] this section outlines the proposal's second research thrust: characterizing the asymptotic behavior of regularity, finding meaningful bounds on multigraded regularity, and generalizing the Eisenbud-Goto correspondence to the toric setting. See Section 1.2 for a brief introduction to toric varietes, in this section we frequently use the following notation.

**Notation 3.1.** *We let* $Y$ *be a smooth projective toric variety coming from a fan* $\Sigma \subset N_{\mathbb{R}}$ *with* $\operatorname{Pic}(Y) \cong \mathbb{Z}^r$*, We denote the* $\operatorname{Pic}(Y)$*-graded Cox ring of* $Y$ *by* $S = \mathbb{C}[x_1, \ldots, x_n]$ *and let* $B \subset S$ *be the irrelevant ideal of* $Y$.

Maclagan and Smith, defined multigraded regularity for coherent sheaves on toric varieties or equivalently $\mathbb{Z}^r$-graded modules over the Cox ring $S$ (Definition 1.5). While it serves the classical role of measuring a module's complexity, multigraded regularity exhibits several new phenomena. For example, $\operatorname{reg}(M)$ is not a number, but a subset of $\mathbb{Z}^r$, and it does not necessarily have a unique minimal element (Figure 3); nor is $\operatorname{reg}(M)$ determined by the multigraded Betti numbers of $M$ [28].

### 3.1 Asymptotic Behavior and Bounds on Multigraded Regularity

In the classical setting of standard graded polynomial rings, significant attention has been given to studying the asymptotic behavior of Castelnuovo–Mumford regularity in various families, e.g., powers of ideals [32, 86, 41, 5] and powers of ideal sheave [39, 40, 42], and providing meaningful geometric bounds on the regularity of ideals defining smooth subvarieties [64, 48, 10, 94]. The same questions for multigraded regularity remain widely open. The PI hopes to resolve this by building upon her recent work [27]

**Research Problem 3.2.** *Using Notation 3.1, understand the asymptotic behavior of* $\operatorname{reg}(I^p)$ *and* $\operatorname{reg}(\mathcal{I}^p)$ *where* $I \subset S$ *is a* $\mathbb{Z}^r$*-graded ideal and* $\mathcal{I} \subset \mathcal{O}_Y$ *is an ideal sheaf.*

Ongoing work of the PI with Lauren Cranton Heller, Mahrud Sayrafi, and Alexandra Seceleanu on led to the following conjecture, which would provide a precise answer to Problem 3.2.

**Conjecture 3.3.** *Let* $Y$ *be a smooth projective toric variety and* $\mathcal{I} \subset \mathcal{O}_Y$ *an ideal sheaf. If* $\pi : \tilde{Y} \to Y$ *be the blow-up of* $Y$ *along* $\mathcal{I}$ *and let* $E$ *denote the exceptional divisor then:*

$$\lim_{p\to\infty} \frac{\operatorname{reg}(\mathcal{I}^p)_{\mathbb{R}}}{p} = \left\{ L \in \operatorname{Pic}(Y)_{\mathbb{R}} \;\middle|\; \pi^*(L) - E \text{ is nef on } \tilde{Y} \right\}$$

*where the limit is in the sense of Painlevé–Kuratowski convergence of subsets of* $\mathbb{R}^n$.

We call the set on the right hand side of Conjecture 3.3 the Seshadri region of $\mathcal{I}$ as it is a natural multigraded analog of Seshadri constants, which measure positivity. One can think of this conjecture as the natural analog of [42]. This conjecture suggests that the asymptotically, multigraded regularity – which is typically complicated and non-convex – is governed by a surprisingly simple, convex geometric object measuring positivity. In ongoing work, the PI has established key steps toward the proof and anticipates resolving this conjecture during the grant period.

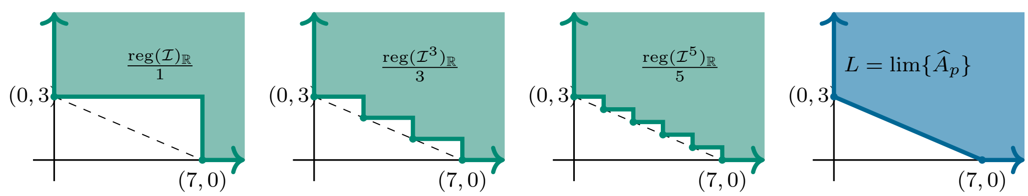


Figure 3. The picture of the convergence of $\frac{\operatorname{reg}(\mathcal{I}^p)_{\mathbb{R}}}{p}$ to the Seshadri region of $\mathcal{I}$ predicted Conjecture 3.3.

Given its definition and that it is a measure of positivity for an ideal sheaf, it is natural to wonder whether the Seshadri region of $\mathcal{I}$ predicted Conjecture 3.3 is connected to the theory of Newton-Okounkov bodies, which are also convex bodies roughly measuring positivity of a linear series along a flag of subvarieties that have received significant study [97, 98, 91, 74, 2, 73]

**Research Problem 3.4.** *Are the Seshadri regions Newton-Okounkov bodies for some flag on* $Y$.

Beyond the asymptotics of $\operatorname{reg}(I^p)$ and $\operatorname{reg}(\mathcal{I}^p)$ it would be useful to have a finer understanding of how these regions change with $p$. Computation and Conjecture 3.3 suggest that the number of minimal generators with respect to the $\operatorname{Nef}(Y)$ ordering of $\operatorname{reg}(I^p)$ and $\operatorname{reg}(\mathcal{I}^p)$ increases with $p$. This growth is a genuinely multigraded phenomenon, as on $\mathbb{P}^n$ regularity is generated by a single element. Quantifying and explaining this growth is an interesting question within reach.

**Research Problem 3.5.** *Using Notation 3.1 let* $I \subset S$ *be a* $\mathbb{Z}^r$*-graded ideal and* $\mathcal{I} \subset \mathcal{O}_Y$ *an ideal sheaf on* $Y$*: (1) Bound the number of minimal generators of* $\operatorname{reg}(I^p)$ *as a function of* $p$*. (2) Bound the number of minimal generators of* $\operatorname{reg}(\mathcal{I}^p)$ *as a function of* $p$.

My hope for part (2) of Problem 3.5 is to refine the convergence in Conjecture 3.3 to produce a lower bound on the number of minimal generators of $\operatorname{reg}(\mathcal{I}^p)$ in terms of the number of lattice points "near" the Seshadri region of $\mathcal{I}$ via Ehrhart theory.

In a different direction, I will prove meaningful bounds on the regularity of the defining ideals of subvarieties of $Y$ in the vein of classical results like Gruson, Lazarsfeld, Peskine [64], Bertram, Ein, Lazarsfeld [10], and the Eisenbud–Goto conjecture [48] (recently shown false [94]).

**Research Problem 3.6.** *Using Notation 3.1 let $Z \subset Y$ be smooth subvariety. Bound the multigraded regularity of $I_Z$ and $S/I_Z$ in terms of* $\operatorname{codim}(Z \subset Y)$ *and the degrees of minimal generators of $I_Z$?*

I will approach Problem 3.6 by beginning with understanding the regularity of $I_Z$ when $Z$ is a torus invariant subvariety of $Y$. In this setting there is a bijection between such subvarieties and cones in the fan defining $X$, and the defining equations of $Z$ are entirely determined by the combinatorics of the corresponding cone.

**Research Problem 3.7.** *Compute the multigraded regularity of torus invariant subvarieties of $Y$ in terms of the combinatorics of the cone defining them.*

Despite being a relatively simple class of subvarieties it has been shown that their multigraded regularity behaves in nuanced ways [27]. The goal for approaching Problem 3.7 is to compute both the higher cohomology of $\mathcal{I}_Z$ and its Hilbert function using the combinatorics of the corresponding cone and working inductively on codimension. A degeneration argument and upper semicontinuity could then be used to extend an answer for Problem 3.7 to an answer for Problem 3.6.

Lastly I will pursue the following focused project concerning the defining equations and regularity of the Deligne-Mumford moduli space of $n$-pointed stable curves of genus zero, denoted by $\overline{M}_{0,n}$, which has rich geometry and combinatorics [85, 56, 79, 72, 84, 93, 30, 59]. Keel and Tevelev constructed an embedding $\Omega_n : \overline{M}_{0,n} \hookrightarrow \mathbb{P}^1 \times \mathbb{P}^2 \times \cdots \times \mathbb{P}^{n-3}$, through which the log canonical embedding given by $\partial\overline{M}_{0,n} + K_{\overline{M}_{0,n}}$ factors [80]. Monin and Rana [95] produced explicit equations, discussed below, which they conjectured defined the scheme-theoretic image of $\Omega_n$. Gillespie, Griffin, and Levinson recently proved this [58].

**Research Problem 3.8.** *Compute the multigraded Castelnuovo–Mumford Regularity and minimal graded free resolution of the defining ideal of $\Omega_n(\overline{M}_{0,n}) \subset \mathbb{P}^1 \times \mathbb{P}^2 \times \cdots \times \mathbb{P}^{n-3}$.*

Using *Macaulay2*, the regularity and free resolution of $\Omega_n(\overline{M}_{0,n})$ is computable for $n = 4, 5, 6, 7$. The Monin-Rana equations defining $\Omega_n(\overline{M}_{0,n})$ arise as follows: Let $x_0^{(i)}, x_1^{(i)}, \ldots, x_i^{(i)}$ denote the coordinates on $\mathbb{P}^i$, then for each $i < j$ let

$$M_{i,j} := \begin{pmatrix} x_0^{(i)}(x_0^{(j)} - x_i^{(j)}) & x_1^{(i)}(x_1^{(j)} - x_i^{(j)}) & x_2^{(i)}(x_2^{(j)} - x_i^{(j)}) & \cdots & x_i^{(i)}(x_{i+1}^{(j)} - x_i^{(j)}) \\ x_0^{(j)} & x_1^{(j)} & x_2^{(j)} & \cdots & x_{i+1}^{(j)} \end{pmatrix}.$$

Define $J_{i,j}$ to be the ideal generated by the $2 \times 2$ minors of $M_{i,j}$. The ideal defining $\Omega_n(\overline{M}_{0,n})$ is then $I_n = \sum_{1 \leq i < j \leq n-3} J_{i,j}$. Each ideal $J_{i,j}$ is independent of $n$ and is resolved by an Eagon-Northcott complex. Further the defining ideals of $I_n$ have natural inductive structure since $I_n = I_{n-1} + J_{1,n-3} + \cdots + J_{n-4,n-3}$. Working inductively and using a mapping cone construction over the short exact sequence arising from ideal sums we can produce a non-minimal free resolution of $\Omega_n(\overline{M}_{0,n})$. The key technical step is to control the multigraded resolutions of intersections $J_{i,j} \cap J_{i',j'}$. If we can control these, we can then hope to minimize the resulting resolution to get the minimal free resolution of $\Omega_n(\overline{M}_{0,n})$. This picture is also compatible with the following alternative approach: $\Omega_n(\overline{M}_{0,n})$ is the image of a product of $GL$-equivariant bundles and so can use Kempf collapsing and the Kempf–Lascoux–Weyman geometric technique for computing syzygies [83, 90, 106], which has recently been applied successfully in similar situations [8]

**3.2 A Generalized Eisenbud–Goto Theorem for Toric Varieties** The relationship between Castelnuovo-Mumford regularity, graded Betti numbers, and linear resolutions is fundamental in classical commutative algebra (see Theorem 1.4 [48]). In the multigraded the analog of a linear resolution is a resolution such that each twist at homological degree $i$ is in $L_i(\mathbf{0})$, i.e., the entries of the differential have total degree one. However, with these definitions the relationship between linear resolutions and regularity breaks down dramatically.

*Example* 3.9. Consider $\mathbb{P}^1 \times \mathbb{P}^2$ so that $S = \mathbb{K}[x_0, x_1, y_0, y_1, y_2]$ and $B = \langle x_0, x_1\rangle \cap \langle y_0, y_1, y_2\rangle$. One checks that $S/B$ is $(0,0)$-regular. The minimal graded free resolution of $(S/B)_{\geq \mathbf{0}} = S/B$ is:

$$S \longleftarrow S(-1,-1)^6 \longleftarrow \begin{matrix} S(-1,-2)^6 \\ \oplus \\ S(-2,-1)^3 \end{matrix} \longleftarrow \begin{matrix} S(-1,-3)^2 \\ \oplus \\ S(-2,-2)^3 \end{matrix} \longleftarrow S(-2,-3) \longleftarrow 0.$$

In particular, $S/B$ does not have a linear resolution as $(-1,-1) \notin L_1(\mathbf{0})$, and so being $\mathbf{d}$-regular does not imply $M_{\geq \mathbf{d}}$ has linear resolution. In alignment with Theorem 1.7 this resolution is $\mathbf{0}$-quasilinear.

Examples, like the one above, led the PI and co-authors, to introduce the weaker notion of quasilinear resolutions (see Definition 1.6) which they used to characterize multigraded regularity on products of projective space [28]. The PI plans to generalize these results to other toric varieties.

**Research Problem 3.10.** *Using Notation 3.1 let $M$ be a $\operatorname{Pic}(Y)$-graded $S$-module. Find the correct generalization of $d$-quasi-linear resolution such that a $\operatorname{Pic}(Y)$-graded $S$-module $M$ is $d$-regular if and only if the truncation $M_{\geq d}$ has a $d$-quasilinear resolution.*

We will begin with the case of Hirzebruch surfaces $\mathcal{H}_t = \mathbb{P}(\mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(t))$ where the PI has the following precise conjecture.

**Conjecture 3.11.** *Let $S$ be the $\mathbb{Z}^2$-graded Cox ring of the Hirzebruch surface $\mathcal{H}_t$. If $M$ is a $\mathbb{Z}^2$-graded $S$-module then $M_{\geq d}$ has a $d$-linear resolution if and only if $d + \operatorname{reg}(S) \subset \operatorname{reg}(M)$.*

The PI's work in [28] suggests a general approach to Problem 3.10 and Conjecture 3.11: i) Find an appropriate collection of line bundles $L_1, \ldots, L_t$ on $Y$ in which one can resolve the diagonal in $\Delta \subset Y \times Y$, ii) Understand the image of these line bundles under the Fourier-Mukai transform whose kernel is this resolution of $\Delta$, iii) Prove strong vanishing results for the higher cohomology of the given line bundles $L_i$. This allows us to construct an explicit resolution of $M_{\geq \mathbf{d}}$ whose graded Betti numbers are given, roughly, by the cohomology of $\tilde{M} \otimes L_i$. Recent work by Hanlon, Hicks, and Lazarev suggest the Bondal-Thomsen collection is a natural choice for the $L_i$; as they give a nice explicit resolution of the diagonal of an arbitrary toric variety and generally have strong vanishing properties [67, 17]. Further, some work has gone into step (ii) for this collection [38]. On Hirzebruch surfaces various collections have been studied even further [19].

While not equivalent to being regular, characterizing when $M_{\geq \mathbf{d}}$ has a linear resolution in the traditional sense is still of interest algebraically.

**Research Problem 3.12.** *Using Notation 3.1 let $M$ be a $\operatorname{Pic}(Y)$-graded $S$-module. Characterize when $M_{\geq \mathbf{d}}$ has a linear resolution in terms of vanishing of the local cohomology groups $H^i_B(M)_t$.*

The PI answered this question for products of projective spaces [28, Theorem 6.2]. Using the same methods of controlling the Betti numbers needed to answer Problem 3.10 would likely provide a way to give a cohomological characterization for when $M_{\geq \mathbf{d}}$ has a linear resolution.

Direct computation of the regularity of a module $M$ is often prohibitively slow due to the complexity of local cohomology. A significantly faster approach involves computing the truncation $M_{\geq \mathbf{d}}$ and its associated minimal free resolution. Consequently, Theorem 1.7, combined with the resolution of Conjecture 3.11 and Problem 3.10, yields a practical algorithm for determining regularity at a single multidegree $\mathbf{d} \in \mathbb{Z}^r$. However, developing an algorithm to compute the entire regularity region of M remains a major open challenge.

**Research Problem 3.13.** *Find an algorithm to compute a minimal generating set for* $\operatorname{reg}(M)$.

This question is open in all cases except $Y = \mathbb{P}^n$. The primary obstacle is determining a box guaranteed to contain all minimal generators of $\operatorname{reg}(M)$. Current bounds typically resemble L-shaped regions (i.e., translates of $\operatorname{Nef}(Y)$), which do not sufficiently constrain the search space for these

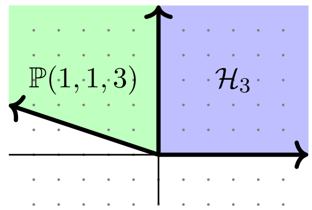


FIGURE 4. The secondary fan of the Hirzebruch surface $\mathcal{H}_3$, which has two chambers; (blue) corresponding to $\mathcal{H}_3$ and (green) corresponding to $\mathbb{P}(1,1,3)$.

generators. Therefore, addressing Problem 3.13 is intrinsically linked to this project's objective of establishing tighter bounds on multigraded regularity.

Finally, as noted above, my previous work shows that the multigraded Betti numbers of a module do not determine its multigraded regularity. One might hope this can be saved by considering the multigraded Betti numbers of all virtual resolutions of $M$. A virtual resolution of a module $M$ is a free $\mathbb{Z}^r$-graded complex of $S$-modules $F_\bullet$ such that the corresponding complex $\tilde{F}_\bullet$ of locally free $\mathcal{O}_Y$-modules is a resolution of the sheaf $\tilde{M}$ [9, Definition 1.1].

**Research Problem 3.14.** *Using Notation 3.1 let $M$ be a* Pic$(Y)$*-graded $S$-module. Does the graded Betti numbers of all virtual resolutions of $M$ determine the multigraded regularity of $M$?*

## 4. Uniform Homological Algebra on Toric Varieties

A distinctive feature of toric geometry, stemming from the GIT presentation $Y = (\mathbb{C}^n \setminus \mathbb{V}(B))/(\mathbb{C}^\times)^r$, is that many non-isomorphic toric varieties share the same graded Cox ring $S = \mathbb{C}[x_1, \ldots, x_n]$. Fixing a $\mathbb{Z}^r$-graded ring $S$ determines the torus action of $(\mathbb{C}^\times)^r$ on $\mathbb{C}^n$, but it does not determine a unique geometric realization; different choices of stability conditions, or equivalently irrelevant ideals, give different projective toric varieties with Cox ring $S$.

The space of all such GIT models is controlled by the secondary (GKZ) fan $\Sigma_{\mathrm{gkz}}$. A choice of a maximal cone (chamber) $\Gamma$ in $\Sigma_{\mathrm{gkz}}$ is equivalent to specifying: i) a toric variety $Y_\Gamma$ with Cox ring $S$, ii) an irrelevant ideal $B_\Gamma \subset S$, or iii) a GIT stability condition determining the semi-stable locus of the quotient. Combinatorially, one may think of $\Sigma_{\mathrm{gkz}}$ as parameterizing the regular sub-division of a given set of rays. Moreover, $\Sigma_{\mathrm{gkz}}$ governs the birational geometry of these models: crossing a wall between adjacent chambers changes stability and the corresponding varieties are related by an explicit birational transformation (flips, flops), as established in VGIT [37]

*Example* 4.1. Consider the case of the Hirzebruch surface $\mathcal{H}_3$. Here the Cox ring is $S = \mathbb{C}[x_1, x_2, x_3, x_4]$ with the $\mathbb{Z}^2$-grading given by $\deg(x_1) = \deg(x_3) = (1,0)$, $\deg(x_4) = (0,1)$, and $\deg(x_2) = (-3,1)$. The secondary/GKZ fan of this Cox ring is shown in Figure 4, and has two chambers one corresponding to $\mathcal{H}_3$ and one corresponding to the weighted projective plane $\mathbb{P}(1,1,3)$. The irrelevant ideals are $B_{\mathcal{H}_3} = \langle x_1, x_3\rangle \cap \langle x_2, x_4\rangle$ and $B_{\mathbb{P}(1,1,3)} = \langle x_1, x_3, x_4\rangle \cap \langle x_2\rangle$.

A developing theme is that many homological properties of toric varieties depend primarily on the graded ring $S$ rather than on a specific model $Y$[76, 77, 66, 6, 67, 7]. Thus, where the classical algebra–geometry dictionary on $\mathbb{P}^n$ fails to generalize, it suggests the need for a dictionary that simultaneously accounts for all VGIT quotients of a fixed ring $S$. These issues are invisible on $|\mathbb{P}^n$ because the secondary fan of $\mathbb{P}^n$ has a single chamber. This project aims to establish a uniform framework for studying homological invariants across different realizations of the same Cox ring.

**4.1 Regularity on the $D_{\text{Cox}}$ Category** A recent instantiation of these ideas is the work of [7] showing that birational variant of King's conjecture on the existence of full strong exceptional collections of line bundles, is true, if one is willing to work on unifying derived category called $D_{\text{Cox}}(Y)$, called the Cox category of $X$. The Cox category of $Y$ is the smallest derived category containing the derived category $D(\mathcal{Y}_i)$ for each toric stack $\mathcal{Y}_i$ corresponding to chambers in the secondary fan of $Y$. While $D_{\text{Cox}}(Y)$ is *not* the derived category of any stack, it has many nice properties. In particular, the Bondal–Thomsen collection is a full strong exceptional collection on $D_{\text{Cox}}(Y)$ and there are natural localization maps from $D_{\text{Cox}}(Y)$ to each VGIT quotient [7].

**Research Problem 4.2.** *Develop a theory of Castelnuovo–Mumford regularity on the category $D_{\text{Cox}}$ that restricts to multigraded regularity on each $Y_i$.*

One natural approach to developing such a theory is to note that $D_{\text{Cox}}(Y)$ can be realized as an explicit full subcategory of $D(\tilde{\mathcal{Y}})$ where $\tilde{\mathcal{Y}}$ is any smooth toric stack with proper birational morphisms to the $\mathcal{Y}_i$. One could thus, hope to using the notion of multigraded regularity on $\tilde{\mathcal{Y}}$ and restrict it to the $D_{\text{Cox}}(Y)$ subcategory. A main obstruction is that the choice of $\tilde{\mathcal{Y}}$ is not unique, and thus, defining regularity in this way may depend on the choice of $\tilde{\mathcal{Y}}$. I hope to circumventing this by showing that there is, in a sense, a best choice for $\tilde{\mathcal{Y}}$; namely the toric stack $\mathcal{Y}_{\text{gkz}}$ constructed in [7, Section 3] by gluing the stack corresponding to each camber of $\Sigma_{\text{gkz}}$ While not unique to $D_{\text{Cox}}(Y)$ this construction of a $\tilde{\mathcal{Y}}$ should be universal in some sense that seems missing from [7, Section 3].

**Research Problem 4.3.** *Characterize the universality of the $\mathcal{Y}_{\text{gkz}}$ stack construction.*

Proving a strong version of universality would provide a strong step towards defining regularity on $D_{\text{Cox}}(Y)$ via $D(\mathcal{Y}_{\text{gkz}})$. As an effort in this direction, and with aim of making explicit some of the steps of the construction provided in [7], I plan to implement this construction in *Macaulay2*.

**Research Problem 4.4.** *Implement the construction of the $\mathcal{X}_{\text{gkz}}$ stack in the toricStacks package being developed for Macaulay2 as part of Problem 2.18.*

Work on this problem is ongoing with Maya Banks and John Cobb. We have rough code that can compute $\mathcal{X}_{\text{gkz}}$ in a narrow number of cases, and we are working to expand its functionality.

**4.2 Homological Invariants and Wall Crossings** An alternative approach of studying homological algebra on all toric varieties with the same Cox ring simultaneously is to understand how things like minimal free resolutions and multigraded regularity change when moving between adjacent chambers $\Gamma$ and $\Gamma'$ of the secondary fan. Geometrically, this corresponds to a birational transformation between the toric varieties $Y_\Gamma$ and $Y_{\Gamma'}$, (e.g. a flip or flop). Algebraically this corresponds to the irrelevant ideal changing $B_\Gamma$ to $B_{\Gamma'}$.

Geometrically we can think of sheaves crossing between $\Gamma$ and $\Gamma'$ by the pushing or pulling along the corresponding birational transformation $Y_\Gamma \to Y_{\Gamma'}$. Algebraically, this roughly corresponds to fixing a $\mathbb{Z}^r$-graded module $M$ on the ring $S$ and then saturating with respect to $B_\Gamma$ or $B_{\Gamma'}$ to get modules $M_\Gamma$ and $M_{\Gamma'}$ corresponding to sheaves on $Y_\Gamma$ and $Y_{\Gamma'}$ respectively. A *wall-crossing formula* seeks to relate the algebraic or geometric invariants of $M_\Gamma$ and $M_{\Gamma'}$. We aim to establish precise wall-crossing formulas for multigraded regularity and multigraded Betti numbers.

**Research Problem 4.5.** *Fix $I \subset S$, a $\mathbb{Z}^r$-graded ideal and $\Gamma$ and $\Gamma'$ be adjacent chambers in $\Sigma_{\text{gkz}}$.*

*(1) (Wall-Crossing for Regularity): Prove there exists a "small" subset $F \subset \mathbb{Z}^r$ independent of $I$ such that:*

$$\operatorname{reg}_{Y_\Gamma}((I :_S B_\Gamma^\infty)) + F = \operatorname{reg}_{Y_{\Gamma'}}((I :_S B_{\Gamma'}^\infty)) + F.$$

*(2) (Wall-Crossing for Betti Numbers): Prove the minimal graded free resolutions of $(I :_S B_\Gamma^\infty)$ and $(I :_S B_{\Gamma'}^\infty)$ agree in small degree.*

Proving these would establish that the homological complexity does not change arbitrarily under birational transformations, but rather follows predictable patterns dictated by VGIT. The strategy is to exploit the combinatorial control one over the irrelevant ideal during a wall-crossing, which corresponds to a bistellar flip that exchanges two minimal primes in the ideals $B_\Gamma$ and $B_{\Gamma'}$ [37, Theorem 15.3.13]. A Mayer–Vietoris type spectral sequence can then relate the local cohomology modules $H^i_{B_\Gamma}((I :_S B_\Gamma^\infty))$ and $H^i_{B_{\Gamma'}}((I :_S B_{\Gamma'}^\infty))$ providing some control over the regularity and minimal free resolutions. This approach is further motivated by the fact these birational maps often induce equivalences on derived categories [75, 76, 77, 66, 78, 6]. As a test case for Problems 4.5, I explore varieties Y with Picard rank two, where the secondary fan is two-dimensional.

## 5. Broader Impacts

**5.1 Organizing** In addition to the 10+ conferences noted in Section 1.5.1, since starting at Dartmouth in 2024 I organized two algebraic geometry conferences: *AGNES* (150 participants) and *BATMOBILE* (50 participants). I am organizing this year's *BATMOBILE*, to occur at Amherst College in November 2025, highlighting recent work in combinatorial algebraic geometry and commutative algebra. I am preparing a BIRS proposal titled *The Geometry of Syzygies: Past & Future* and plan to organize the next *Gems of Commutative Algebra* at Dartmouth in Winter 2027 to promote community among the next generation of researchers; if successful, I will write a short Notices of the AMS article on these efforts and how to organize similar conferences amid current logistical hurdles. I am an organizer of Dartmouth's *Algebra & Number Theory* seminar.

**5.2 Campus Community.** In an effort to promote community among math majors on Dartmouth's campus, I serve as the faculty advisor to the college's *AWM Student Chapter* and am also working with the campus's *oSTEM* chapter to broaden participation in STEM. Beginning in Fall 2025 I am organizing a graduate student seminar *Unwritten Rules* providing professional development to graduate students in the department facing additional career challenges.

**5.3 National & International Advocacy.** I have worked with the *AWM* Executive Committee to expand support for broader groups of mathematicians and plan to continue this effort. From Winter 2023 to Spring 2025 I served on SLMath's *Committee on Women in Mathematics* until its dissolution. I have since agreed to serve on two *AMS* committees to promote mathematics and the community. I hope to resume work with *Spectra*, where I previously served as inaugural president.

**5.4 Mentoring.** I am mentoring two graduate students, Bailee Z. (Michigan) and Jacob B. (KSU), on a tropical geometry project begun at MREG 2023; a preprint is expected soon. Since Fall 2024, I've mentored two undergraduates: Joanna S. (algebra reading course) and Lisa S. (exploring math graduate school). Together with Asher Auel I am co-advising one graduate student, Will D., working on using free resolutions of points on surfaces to study the geometry of the moduli of spin curves. I am pre-advising three students (Allison T., Alex M., Aidan H.) interested in combinatorics/algebra. With these students I am developing a reading course in algebraic geometry aimed at increasing retention among students who are often pushed away from the field, by taking a humanistic approach, combining active learning, exercises in algebraic geometry, and approaches to community building. I will make the notes I am writing for for this available on my website. I expect to become the advisor for at least one of these students. I've also agreed to lead an undergraduate summer research project at *Count Me In* (organized in part by Milena Hering and Diane Maclagan).

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